Lecture 16: Heat Engines and the Second Law

CBE 20260 / Thermodynamics I

Edward Maginn

Table of contents

1 Introduction: Heat vs. Work

So far, we have covered energy and entropy balances for various pieces of process equipment (pumps, compressors, turbines, etc.). Now we want to "put it all together" to produce **Work**.

Why focus on Work? While Heat is useful (warming our homes), **Work** is especially valuable because it is an organized form of energy transfer. We use it to move things (cars, planes), generate electricity, and drive chemical processes.

There is a fundamental asymmetry between heat and work:

1. **Work** $\rightarrow$ **Heat:** Easy. We do this all the time via friction (rubbing hands together), electrical resistance, or Joule's paddlewheel experiment. It is 100% efficient.
2. **Heat** $\rightarrow$ **Work:** Hard. We cannot simply convert 100% of heat into work.

To understand why, we will analyze abstract "devices" that attempt these conversions.

2 The Kelvin-Planck Device

Let's imagine a steady-state cyclic device that takes in heat (Q) from a single reservoir and converts it entirely into work (W). We call this a **Kelvin-Planck Device**.

2.1 Analysis

Consider this device operating in a cycle (or steady state):

First Law:

$$\Delta U = 0 = Q_{in} + W_{out}$$
$$W_{out} = Q_{in}$$

According to the First Law, this device is **possible**. Energy is conserved.

Second Law:

$$\frac{dS_{sys}}{dt} = \sum \frac{\dot{Q}}{T_b} + \dot{S}_{gen}$$

At steady state $(dS/dt = 0)$:

$$0 = \frac{\dot{Q}_{in}}{T} + \dot{S}_{gen}$$
$$\dot{S}_{gen} = -\frac{\dot{Q}_{in}}{T}$$

Since $Q_{in} > 0$ (heat added) and $T > 0$, this implies:

$$\dot{S}_{gen} < 0$$

This violates the Second Law! ($\dot{S}_{gen}$ must be ≥ 0).

! Kelvin-Planck Statement of the Second Law

It is impossible to construct a device that operates in a cycle and produces no other effect than the production of work and transfer of heat from a single body. (*i.e., You cannot turn Heat 100% into Work*).

2.2 The Reverse Kelvin-Planck Device

What if we reverse the arrows? A device that takes Work and turns it into Heat?

- **1st Law:** $Q_{out} = W_{in}$. (Possible)
- **2nd Law:** $0 = \frac{-\dot{Q}_{out}}{T} + \dot{S}_{gen} \implies \dot{S}_{gen} = \frac{\dot{Q}_{out}}{T} > 0$.

This is **Possible**. This is simply "friction" or electrical heating. Work can be converted 100% to Heat.

3 The Clausius Device

Now consider a device that transfers heat from a Cold reservoir (T_C) to a Hot reservoir (T_H) without any work input.

First Law: $Q_{in} = Q_{out}$ (Energy conserved). **Second Law:**

$$0 = \frac{Q_C}{T_C} - \frac{Q_H}{T_H} + S_{gen}$$

Since $Q_C = Q_H = Q$:

$$S_{gen} = Q\left(\frac{1}{T_H} - \frac{1}{T_C}\right)$$

Since $T_H > T_C$, the term $(\frac{1}{T_H} - \frac{1}{T_C})$ is **negative**. Therefore, $S_{gen} < 0$. **Impossible.** Heat cannot spontaneously flow from Cold to Hot.

> **!** Clausius Statement of the Second Law
>
> It is impossible to construct a device that operates in a cycle and produces no other effect than the transfer of heat from a lower-temperature body to a higher-temperature body.

3.1 The Reverse Clausius Device

If we flow heat from Hot (T_H) to Cold (T_C):

$$S_{gen} = Q\left(\frac{1}{T_C} - \frac{1}{T_H}\right)$$

Since $T_H > T_C$, the term is positive. $S_{gen} > 0$. **Possible.** Heat naturally flows from Hot to Cold.

4 The Heat Engine

We have established two facts: 1. We cannot convert Heat 100% to Work ($S_{gen} < 0$). 2. Heat naturally flows from T_H to T_C ($S_{gen} > 0$).

The Engineering Idea: What if we combine these? What if we take the natural flow of heat from T_H to T_C (which generates entropy) and "bleed off" some of that energy as Work?

Let's analyze this **Heat Engine**: * Heat Q_H enters from Hot Reservoir T_H. * Heat Q_C leaves to Cold Reservoir T_C. * Work W is produced.

4.1 First Law Analysis

$$\Delta U = 0 = Q_H - Q_C - W$$
$$W = Q_H - Q_C$$

(*Note: Here we define Q_C as a positive magnitude leaving the system*).

4.2 Second Law Analysis

$$0 = \frac{Q_H}{T_H} - \frac{Q_C}{T_C} + S_{gen}$$

Rearranging to solve for Q_C:

$$\frac{Q_C}{T_C} = \frac{Q_H}{T_H} + S_{gen}$$

$$Q_C = T_C \left(\frac{Q_H}{T_H} + S_{gen} \right)$$

$$Q_C = Q_H \frac{T_C}{T_H} + T_C S_{gen}$$

4.3 Combining First and Second Laws

Substitute the expression for Q_C back into the Work equation ($W = Q_H - Q_C$):

$$W = Q_H - \left(Q_H \frac{T_C}{T_H} + T_C S_{gen} \right)$$

$$W = Q_H \left(1 - \frac{T_C}{T_H} \right) - T_C S_{gen} \tag{1}$$

This is a **remarkable result**. It tells us exactly how much Work we can get out of a heat engine.

1. **The Potential:** The term $Q_H(1 - T_C/T_H)$ represents the *maximum possible work* allowed by the laws of thermodynamics (when $S_{gen} = 0$).
2. **The Penalty:** The term $T_C S_{gen}$ reduces the work output.

5 Thermal Efficiency

We define the thermal efficiency η as:

$$\eta = \frac{\text{Benefit}}{\text{Cost}} = \frac{W}{Q_H}$$

Using our derived expression for Work:

$$\eta = \frac{Q_H(1 - T_C/T_H) - T_C S_{gen}}{Q_H}$$

$$\eta = \left(1 - \frac{T_C}{T_H} \right) - \frac{T_C S_{gen}}{Q_H}$$

5.1 Maximum (Reversible) Efficiency

If the process is ideal (Reversible, $S_{gen} = 0$), we get the **Carnot Efficiency**:

$$\eta_{max} = 1 - \frac{T_C}{T_H}$$

To maximize efficiency, we want: * T_H to be as **High** as possible. * T_C to be as **Low** as possible.

This explains why power plants use superheated steam (high T_H) and cooling towers/rivers (low T_C).

6 Lost Work

Real engines always have irreversibilities (friction, heat loss, turbulence), so $S_{gen} > 0$. The difference between the Maximum Work and the Actual Work is called **Lost Work**.

$$W_{lost} = W_{max} - W_{actual}$$
$$W_{lost} = \left[Q_H\left(1 - \frac{T_C}{T_H}\right)\right] - \left[Q_H\left(1 - \frac{T_C}{T_H}\right) - T_C S_{gen}\right]$$

$$W_{lost} = T_C S_{gen}$$

This simple equation links the abstract concept of entropy generation directly to lost engineering potential. Every unit of entropy generated destroys our ability to do useful work by an amount proportional to the sink temperature T_C.